

TaskifyAI

Recommended Tech Stack

Full-Stack Architecture Guide · March 2026 · v1.0

5	15+	Next.js 15	Supabase	Vercel
Layers	Technologies	Framework	Database	Hosting

The following stack is purpose-built for TaskifyAI's core requirements: **dark-mode-first UI with rich animations, real-time AI streaming, real-time task sync, and gamification logic.** Every tool was chosen for developer ergonomics, tight Next.js integration, and minimal operational overhead.

Frontend

User interface, state management & animations

Next.js 15

FRAMEWORK

App Router + RSC

SSR / SSG / ISR / Edge

TypeScript

LANGUAGE

End-to-end type safety

Strict mode enabled

Tailwind CSS

STYLING

Dark mode first

Design token mapping

shadcn/ui

COMPONENTS

Accessible primitives

Customisable variants

Framer Motion

ANIMATION

Spring physics

Task completion FX

Zustand

STATE

Task + XP store

Lightweight & simple

React Hook Form

FORMS

Zod validation

Task creation flows

Recharts

DATA VIZ

Analytics dashboard

Streak / progress graphs

Next.js 15	App Router gives you Server Components for fast initial loads + Client Components for interactive animations. Server Actions eliminate a separate API layer for mutations like creating/completing tasks.
Tailwind + shadcn	The design doc calls for a precise dark-mode-first token system. Tailwind maps 1:1 to the CSS variables defined in the design tokens. shadcn gives unstyled accessible primitives you fully own — no fighting a component library's opinions.
Framer Motion	The design specifies spring physics for priority re-sorting (layout animation), confetti on task completion, and drag-and-drop reordering. Framer Motion handles all three natively and integrates seamlessly with React.
Zustand	Task state, user XP, streak counters, and UI state (sidebar open/closed, active filters) need to be globally accessible. Zustand is 1KB, has no boilerplate, and supports Immer-style mutations.

API Layer

Server logic, AI integration & authentication

Next.js API Routes

BACKEND

REST endpoints
Server Actions for mutations

Vercel AI SDK

AI STREAMING

Streaming responses
Tool calling support

NextAuth.js v5

AUTH

OAuth + magic link
JWT sessions

Zod

VALIDATION

Schema validation
Type-safe API contracts

Vercel AI SDK

Built specifically for streaming AI responses in Next.js. Handles SSE streaming, tool-calling, and multi-modal inputs out of the box. The useChat / useCompletion hooks make the typewriter AI effect trivial to implement.

NextAuth.js v5

Integrates natively with Next.js App Router middleware. Supports Google/GitHub OAuth + passwordless magic-link email — exactly the low-friction auth flow the design calls for. v5 is purpose-built for the App Router.

Zod

Validates all incoming API payloads and task creation forms. Shared schemas between frontend and backend eliminate a whole class of type-mismatch bugs. Works seamlessly with React Hook Form.

AI Services

Language models, embeddings & smart features

GPT-4o

LANGUAGE MODEL

Task parsing + briefings
Smart scheduling

text-embedding-3

EMBEDDINGS

pgvector semantic search
Task grouping

Whisper API

VOICE (PHASE 2)

Voice task creation
Audio transcription

GPT-4o

Powers every AI feature: natural language task creation via the Command Bar, daily briefing card generation, priority suggestions with reasoning tooltips, blocker detection, and the task rewrite/polish feature. GPT-4o's speed makes streaming feel snappy.

text-embedding-3

Converts tasks into vector embeddings stored in pgvector (built into Supabase). Enables semantic search ('find all tasks about the project proposal'), smart task grouping, and future AI-powered recurring task detection.

Whisper (Phase 2)

The design specifies a voice input mode in the Command Bar. Whisper transcribes audio, GPT-4o parses the intent, and the task is created — all in one pipeline via the Vercel AI SDK.

Database & Storage

Persistence, caching & real-time sync

Supabase

DATABASE

PostgreSQL + pgvector
Realtime subscriptions

Supabase Auth

AUTH STORAGE

User records + RLS
Session management

Upstash Redis

CACHE

Session + API cache
AI rate limiting

Supabase Storage

FILES

User avatars
Exported reports (Phase 2)

Supabase	Single platform for PostgreSQL, Auth, Storage, and Realtime. The realtime subscriptions allow live task sync across multiple browser tabs — if you complete a task on mobile, your desktop updates instantly. Row Level Security ensures users only access their own data. pgvector is built-in for embeddings.
Upstash Redis	Serverless Redis with per-request pricing — perfect for Vercel. Used for: caching AI responses (avoid re-generating the same daily briefing), rate-limiting the AI endpoint per user, and storing ephemeral UI state like 'last visited date' for streak calculation.

Infrastructure & Tooling

Deployment, monitoring & developer workflow

Vercel HOSTING / CDN Zero-config Next.js Edge functions + CDN	GitHub Actions CI/CD Auto deploy on push Type-check + lint	Sentry ERROR TRACKING Frontend + API errors Performance tracing	PostHog ANALYTICS Completion rates Feature flag support	Resend EMAIL Streak reminders Magic link delivery
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Vercel	Native deployment platform for Next.js. Automatic preview deployments on every PR, edge functions for low-latency AI streaming, and built-in CDN for static assets. Free tier is generous enough for MVP.
PostHog	Tracks the exact metrics in the PRD: DAU/MAU, task completion rate, streak unlock rate, and session duration. Feature flags let you roll out AI features incrementally without redeployments.
Resend	Modern email API that pairs with React Email for templated emails. Used for magic-link auth delivery and optional streak reminder notifications — two of the key engagement drivers in the design.

Phase rollout

Phase 1 (Weeks 1–6)	Next.js 15 · TypeScript · Tailwind · shadcn/ui · Framer Motion · Zustand · Supabase (DB + Auth) · NextAuth.js · GPT-4o via Vercel AI SDK · Vercel hosting
Phase 2 (Weeks 7–12)	Redis (Upstash) · text-embedding-3 + pgvector · Whisper voice input · PostHog feature flags · Resend streak emails · GitHub Actions CI/CD · Sentry monitoring